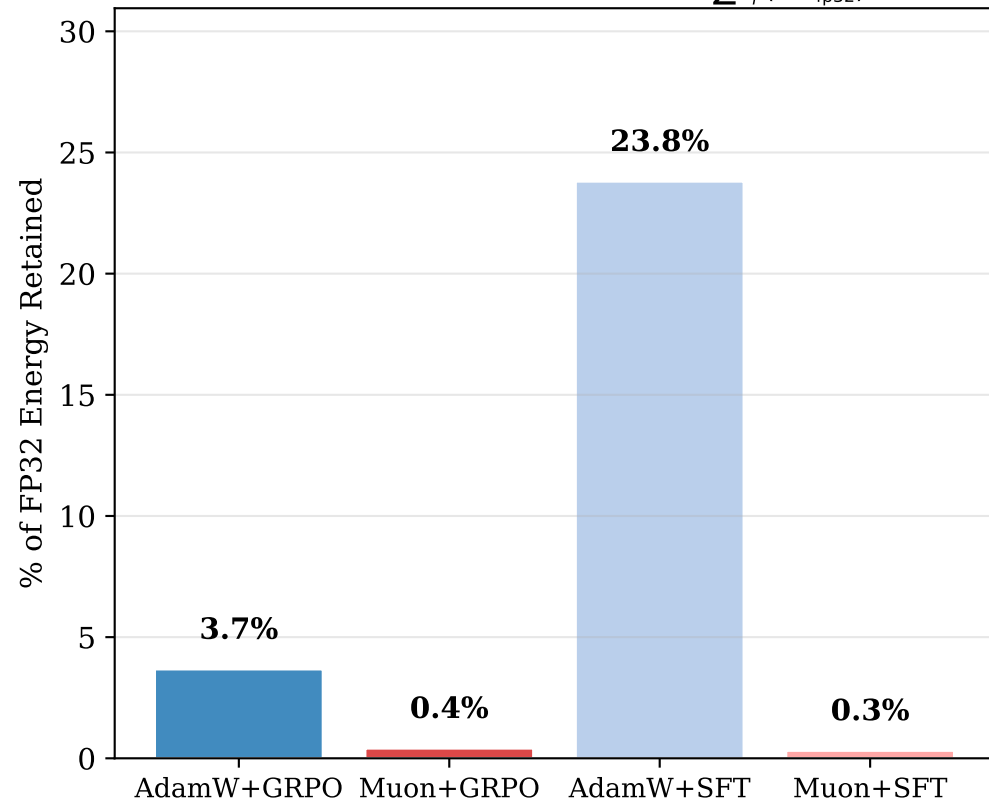


How Much Spectral Mass Survives BF16 Training?

Muon loses 96-99.7% of spectral energy; AdamW retains more because its top SVs survive rounding

Energy Retained in BF16: $\frac{\sum \sigma_i^2(\Delta W_{\text{bf16}})}{\sum \sigma_i^2(\Delta W_{\text{fp32}})}$



Nuclear Norm Retained in BF16: $\frac{\sum \sigma_i(\Delta W_{\text{bf16}})}{\sum \sigma_i(\Delta W_{\text{fp32}})}$

